

IndiaMART — Data Requirements

Every data field needed to implement the 5-phase churn reduction system — what exists, what needs building, and what's external.

18 fields already exist in IM systems	6 need new pipelines or logic	3 external data sources (nice-to-have)	5 computed / derived features
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DATA REQUIREMENTS BY PHASE

PHASE 1 Onboarding Health Check		Day 0–30 6 fields
mcats ID <i>seller_master.mcat_id</i> [P1] [P2] [P3]	Exists	Primary product category. Used to query buyer demand index and peer BL benchmarks.
City / City Tier <i>seller_master.city + city_tier_map</i> [P1] [P2] [P3]	Exists	Seller's registered city and tier (1/2/3). Drives hyperlocal demand check and peer clustering.
Buyer Search Volume <i>buyer_search_index.daily_counts</i> [P1] [P3]	Needs pipeline	How many buyers searched this mcat in this city in the last 30 days. <i>Action: aggregate buyer_search_logs → write to mcat_city_demand_index daily. 1-day build.</i>
NSD Verification <i>nsd_verification.status</i> [P1]	Exists	Is the business NSD-verified? Unverified sellers have 3× higher 60-day churn.
Peer BL Avg (mcats+city) <i>COMPUTED from bl_transactions</i> [P1] [P2] [P3]	Computed	Median BL count/month for all active sellers in same mcat + city tier. <i>Computed nightly via SQL window function on bl_transactions. Written to seller_churn_features table.</i>
First BL Response Time <i>bl_transactions.first_response_ts</i> [P1]	Needs indexing	Time delta between BL received and seller's first response. Best single onboarding predictor. <i>Action: add indexed computed column to bl_transactions. Half-day build.</i>
PHASE 2 Churn Predictor — Model Features		Daily scoring 9 fields
BL Count (30/60/90d) <i>bl_transactions.count [rolling]</i> [P2]	Exists	Rolling BL count per seller over 3 windows. Core volume feature for the model. <i>Action: rolling window query in nightly feature job.</i>
BL Velocity Drop % <i>COMPUTED daily</i> [P2]	Computed	$(BL\ last\ 30d - BL\ prev\ 30d) / BL\ prev\ 30d$. Strongest single churn signal in the model.

DAU / Last Login Date <i>seller_sessions.last_active_ts</i> [P2] [P3]	Exists Days since last platform login. Seller inactive 22+ days = red flag. Simple but powerful.
LMS Response Rate <i>lms_activity.response_rate</i> [P2] [P3]	Exists % of received BLs the seller responded to in last 30 days. Declining rate = disengagement.
PNS Call Count <i>pns_calls.call_count</i> [P2]	Exists Number of PNS/support calls in last 30 days. High call count = high frustration signal.
PNS Call Sentiment NEW — Whisper + LLM on call recordings [P2] [P3]	New pipeline Sentiment score (positive / neutral / frustrated) from AI transcription of call recordings. <i>Action: Whisper transcription → LLM classifier → write to pns_call_insights table. 1–2 week build.</i>
Peer BL Delta COMPUTED daily [P2] [P3]	Computed Seller's BL count vs median of same mcat + city cluster. Gap > 50% = high churn risk.
Days to Renewal <i>subscription_master.expiry_date</i> [P2] [P4] [P5]	Exists Days until subscription expires. Critical for timing every intervention window.
Historical Churn Label <i>subscription_master.renewed_flag</i> [P2]	Exists Did seller renew within 30 days of expiry? Training label for LightGBM model. Needs 12 months.

PHASE 3

Retention — Nudge + Dashboard + Winback

1–2 months before dropout | 8 fields

Seller Language / State <i>seller_master.state + language_map</i> [P3]	Exists Registration state → maps to regional language (Hindi, Gujarati, Tamil...) for WhatsApp messages.
WhatsApp Opt-in Status <i>seller_comms.whatsapp_optin</i> [P3]	Check coverage Has seller opted in to WhatsApp comms? Required gate for WhatsApp Business API delivery. <i>Action: audit current opt-in coverage — may need re-consent campaign for gaps.</i>
Top Peer Seller Data COMPUTED weekly from bl_transactions [P3]	Computed Top-5 sellers in same mcat + city tier by BL count — anonymised. Used for dashboard comparison card. <i>Weekly aggregation job. Output to peer_benchmark table.</i>
Profile Completeness <i>seller_profile.completeness_pct</i> [P3]	Exists % of profile filled — photos, description, products, certifications. Gap vs peer = key message.

Churn Reason (Classified) NEW — LLM on call transcripts + LMS notes [P3]	New pipeline Classified reason: No BL / Price / Competitor / Engagement / Personal. Action: LLM classification on transcribed calls + CRM agent notes. Routes winback script to correct track.
Gifted Lead Pool buyer_leads [filtered + reserved] [P3] [P5]	Needs logic Verified buyer leads: posted < 72hr, buyer has past purchase history, < 3 sellers assigned. Action: build gifted_lead_reservations table + allocation microservice. 3–4 day build.
Past BL Conversion bl_transactions.conversion_flag [P3]	Reliability check Did seller mark past BLs as converted? Used to estimate ROI from IM. Note: self-reported field — reliability varies. Cross-check with LMS response patterns.
Winback CRM History winback_crm.attempts + outcomes [P3]	Exists Previous winback call attempts, outcomes, agent notes. Prevents duplicate outreach. Action: structured export needed — CRM data may be in free-text notes.
PHASE 4 + 5 BL Upgrade + Renewal Window Boost 10–15d before dropout -7d to +7d renewal window 5 fields	
Current Subscription Tier subscription_master.plan_id [P4] [P5]	Exists Seller's current plan (Bronze/Silver/Gold/Platinum). Determines which tier's BLs to unlock.
BL Tier Mapping bl_tier_config.plan_bl_map [P4]	Exists Which BL quality / order-value range belongs to which subscription tier.
Renewal Date (Exact) subscription_master.expiry_date [P4] [P5]	Exists Exact renewal date to trigger the -7d and +7d boost windows. Reused from Phase 2.
Premium BL Pool buyer_leads [filtered + reserved] [P4] [P5]	New logic High-value BLs: order value above threshold, recurring buyer history, posted < 72hr. Action: extend gifted_lead_reservations logic for boost-window pool. Separate from Phase 3 gifted pool.
Post-Renewal Engagement seller_sessions + lms_activity [P5]	Needs tagging Did seller log in and consume BLs in 7 days post-renewal? Measures if boost worked. Action: add event tag 'post_renewal_window' to session and LMS events for 7 days after renewal. Feeds model.

WHAT NEEDS TO BE BUILT — RANKED BY EFFORT

Low effort**Daily aggregation jobs on existing tables**

What: Rolling BL count (30/60/90d), BL velocity drop %, peer BL avg per mcat+city, peer BL delta per seller.

How: Scheduled SQL jobs (Airflow DAG) running nightly on existing bl_transactions table. Results written to seller_churn_features table.

Time: 1–2 days. No new data collection. Pure SQL engineering.

Build effort



1%

Low effort**Buyer search volume aggregation pipeline**

What: Daily count of buyer searches per mcat per city from IM's existing search logs.

How: Aggregate buyer_search_logs → write to mcat_city_demand_index table. Used in onboarding check and nudge message personalisation.

Time: 1 day. Data already exists.

Build effort



1%

Medium effort**Gifted lead pool — reservation + matching logic**

What: Filter buyer_leads for: posted < 72hr · buyer has past purchase history · order value ≥ threshold · < 3 sellers already assigned. Reserve 1 lead per at-risk seller.

How: New microservice + gifted_lead_reservations table to track allocations and prevent double-assignment. Quality gate is non-negotiable.

Time: 3–4 days. Involves allocation logic and quality gates.

Build effort



40%

Medium effort**WhatsApp Business API — personalised message generator**

What: Message factory assembling seller-specific WhatsApp messages from live variables — name, language, buyer search count, peer BL gap, days to renewal.

How: Python Jinja2 template engine + WhatsApp Business API integration. Language mapping table (state → language) needed.

Time: 3–5 days including API integration + Meta template approval process.

Build effort



40%

High effort**PNS call transcription + sentiment + reason classification pipeline**

What: Record → Whisper/Deepgram transcription → LLM sentiment classification (positive/neutral/frustrated) → churn reason extraction → CRM auto-summary.

How: Integrate with PNS call recording system. Run Whisper post-call. LLM classifies: No BL / Price / Competitor / Engagement / Personal. 3-line summary auto-logged to CRM. Rep touches nothing.

Time: 1–2 weeks. Needs call recording access, transcription infra, LLM API, CRM write-back.

Build effort



70%

High effort

LightGBM churn model — training + daily serving pipeline

What: Train on 12 months of churn labels + 7 features. Daily batch scoring. SHAP explanation per seller. Output to seller_churn_scores table.

How: Python LightGBM + SHAP library. Training on historical data. Daily inference job. Feature store for consistent daily features. Monthly retraining.

Time: 1–2 weeks. Feature engineering pipeline is the longest part.

Build effort



80

EXTERNAL DATA — NICE TO HAVE, NOT BLOCKING

GST Filing Activity

Is the seller's business actively filing GST returns? Stronger viability signal than NSD alone — a lapsed GST filer is almost certainly winding down.

Source: GSTN API. Requires seller consent. Adds ~8% precision to onboarding risk score.

Competitor Platform Activity

Is a churned seller now active on JustDial / TradeIndia / Alibaba India? Informs winback script — 'Competitor' route becomes much more targeted.

Source: third-party web crawl or seller self-report on winback call. Enriches churn reason classification.

Commodity / Raw Material Prices

For sellers in metal, textile, chemical categories — input price shocks correlate with subscription cancellation spikes 30–60 days later.

Source: MCX / CRISIL public indices. Adds early-warning signal for category-level churn clusters.

MINIMUM VIABLE DATASET — FOR HACKATHON DEMO

You don't need all 27 fields to demonstrate the system working. The following 5 tables are sufficient to train a basic churn model, generate peer comparisons, show the renewal window graph with real numbers, and simulate the gifted lead flow:

Table	History	What it enables
bl_transactions	30 days	BL count, velocity drop, peer delta, first response time
seller_sessions	30 days	DAU, days since last login
subscription_master	12 months	Renewal dates, plan tier, historical churn labels
seller_master	Full	mcat, city, state, NSD status
lms_activity	30 days	BL response rate — model feature + dashboard gap card